

Introduction to Retrieval-Augmented Generation (RAG)

What is RAG?

Retrieval-Augmented Generation (RAG) is an AI approach that combines information retrieval with a language model. Instead of relying only on the model's learned knowledge, RAG retrieves relevant information from a document collection and uses that information to generate a grounded answer.

Document Processing

A RAG system usually receives documents such as PDF, TXT, or Markdown files. The text is extracted, cleaned, and divided into smaller pieces called chunks.

Chunking

Chunking divides a long document into smaller sections. Good chunking helps the system retrieve focused and relevant information when a user asks a question.

Embeddings

An embedding converts text into a numerical vector that represents its semantic meaning. Similar pieces of text have vectors that are close to each other in vector space.

Vector Store

A vector store keeps document embeddings and allows the application to search for chunks that are semantically similar to a user's query.

Retrieval

When a user asks a question, the query is converted into an embedding. The system searches the vector store and retrieves the most relevant document chunks.

Generation

The retrieved chunks are provided as context to a language model. The model uses that context to produce an answer that is grounded in the available documents.

Benefits

RAG can improve factual grounding, make private or organization-specific information searchable, and allow knowledge to be updated by adding new documents without retraining the language model.